



Environmental Problems, Their Causes, and Sustainability

Md Ekhtekharul Islam

Adjunct Faculty, East West University

Email: Ekhtekharul.islam@ewubd.edu WhatsApp: 01730427425)

By the end, you can:

- Define an environmentally sustainable society
- Explain ecological footprint + Earth Overshoot Day
- Describe pollution types + the prevention-first approach
- Use IPAT + exponential growth to explain “why problems happen”
- Connect ideas to real cases: Bangladesh RMG, China, fast fashion, AI data centres

[1-1 Sustainable Society](#)

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[Individuals Matter \(Leopold\)](#)

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THE SUCCESS STORY

The industry that changed Bangladesh

- \$39.3 billion of garment exports in FY2024–25 — about 81% of everything the country sells abroad
- Roughly 4 million workers, the majority of them women (BGMEA)
- 287 LEED-certified green factories, 124 of them Platinum — more than any other country on earth (May 2026)

THE BILL

What the same decade cost

- Wet processing uses about 300 litres of water per kg of fabric — roughly 6× international best practice
- The sector draws around 1,500 billion litres of groundwater a year — comparable to all of Dhaka's residents
- Dhaka's water table is falling 2–3 metres every year
- Dye-coloured effluent still reaches the Turag and the Buriganga

Both columns are true, and they describe the same factories. So — is this development?

You are on the committee. You have 3 minutes.

**A European buyer offers your factory in Gazipur its biggest order ever.
One condition: a 20% lower unit price.**

OPTION 1 — Say yes today

Run the existing dyeing line. 300 litres of water per kg.
Untreated discharge to the canal. Order confirmed this week.

OPTION 2 — Fix the line first

Install effluent treatment and water recycling before you accept.
It costs money now, and the buyer may go to Vietnam while you install.

*3,000 jobs in your area depend on this factory.
The canal runs past four villages before it reaches the Turag.*

Vote first — no discussion

- A) Take the order as it is
- B) Install first, risk the order
- C) Take it now, invest next year
- D) Refuse these terms

Then argue it out (2 min):

1. Who gains, who pays — and when?
2. Is “we need the jobs” a complete answer?
3. What would make the clean option the cheap option?

What one Bangladeshi factory actually did

DBL Group's Hamza Textiles faced this exact trade-off — and chose to fix the line.

300 → 60

litres of water
per kg of fabric

– 25%

electricity used
per unit of output

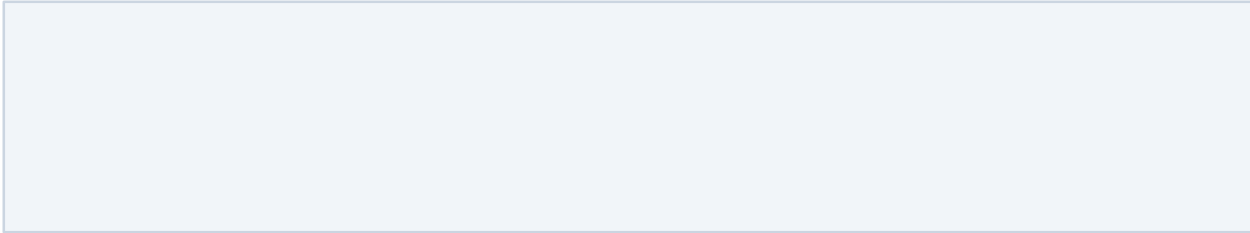
\$80k → \$500k

invested once → saved
every single year

The equipment paid for itself in under three months. The canal got cleaner as a side-effect of a decision the accountants would have made anyway.

**Sustainability is not the opposite of growth.
It is the only version of growth that can keep going.**

1 minute — write a 10-word definition:



Then: share with a partner

→ combine into ONE best definition

Quick vote:

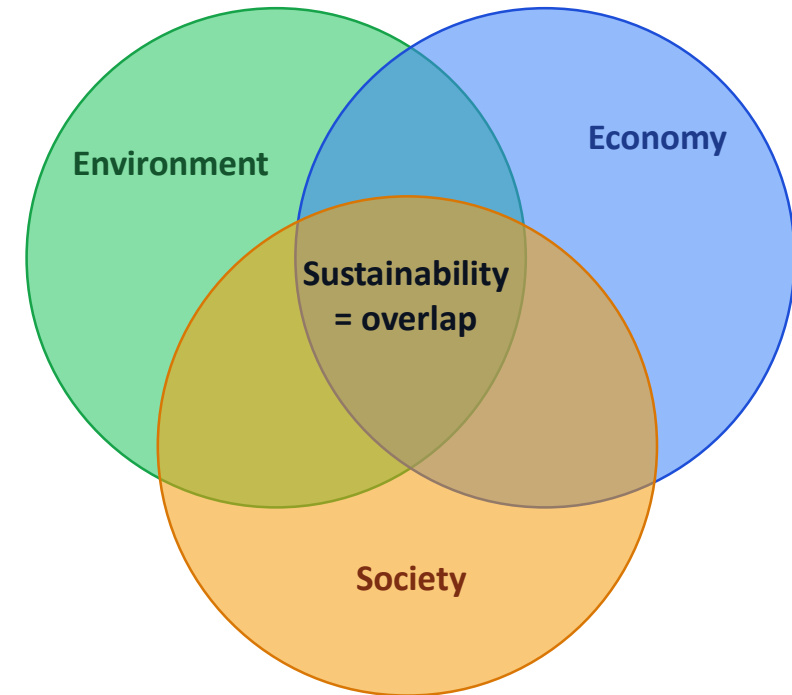
Which feels most urgent?

- A) Air
- B) Water
- C) Waste
- D) Climate

Working definition (classic):

“Meets the needs of the present without compromising the ability of future generations to meet their own needs.”

The “3 circles” (simple):



Natural capital, human impacts, and solutions

1. Natural Capital

Natural resources



Inexhaustible
solar energy



Renewable
forests, water,
fish, soil



Nonrenewable
oil, gas, coal,
metals

Ecosystem services



clean air
& water



climate
regulation



soil
protection



nutrient
cycling



Supports life and human economies

2. Human Degradation



Overuse of
renewables



Pollution
& waste



Forest
conversion



Persistent
plastics &
chemicals



Degraded ecosystems and reduced services

3. Sustainability Solutions



Protect
ecosystems



Use resources within
sustainable yield



Reduce
pollution



Educate & act
together



Maintain natural capital
for future generations



Sustainability depends on conserving natural capital, reducing human damage, and applying collective solutions.

Natural capital: the “invisible” services we rely on



Ecosystem services (examples):

- Clean air + water
- Pollination
- Soil formation
- Climate regulation
- Flood protection
- Recreation + mental health

Think-pair-share (2 min):
Pick one service above.
How does it support the economy
or human health?

Natural Capital

Natural Capital = Natural Resources + Ecosystem Services

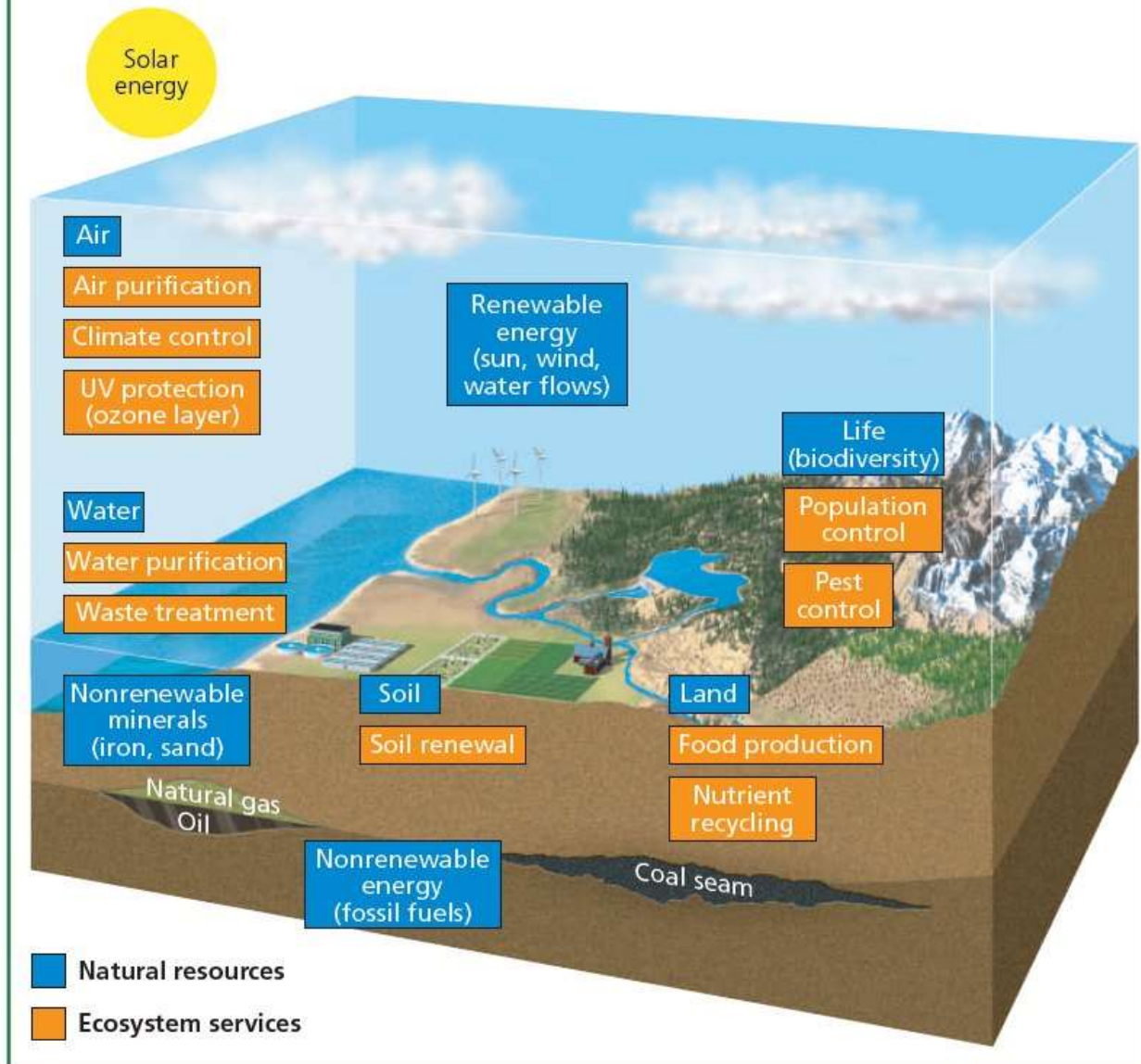


FIGURE 1.3 Natural Capital consists of natural resources (blue) and ecosystem services (orange) that support and sustain the earth's life and human economies (**Concept 1.1B**).

Renewable and Nonrenewable Resources



Provisioning	Regulating	Cultural
Products humans obtain from ecosystems: • Food • Raw Materials e.g. wood, fuel, fibre • Medicine • Fresh Water	Services nature provides that regulate the environment: • Air Quality • Climate • Water Purification • Waste Treatment • Disease and Pest Control • Pollination • Extreme Events Moderation	Non-material benefits of nature for humans: • Recreation e.g. tourism • Aesthetic Values • Religious and Spiritual Values • Mental and Physical Health • Education
Supporting		
The underpinning services that enable all other services to function – encompasses both human and ecosystem needs: • Photosynthesis • Nutrient Cycling • Soil Formation		

RIVERS
FORESTS
AIR
STREAMS

Open-access resources

Shared resources

Owned by everyone



Water, which has been colored by textile dye and will eventually flow through the Turag and Buriganga rivers, is released near a paddy field in the Mawna Union area, north of Gazipur. [Source: Reuters]



The Tragedy of the Commons

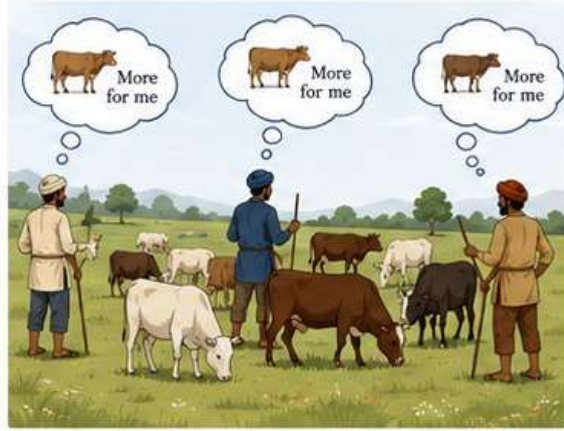
When individuals acting in self-interest overuse a shared resource, everyone loses.

1 Shared resource



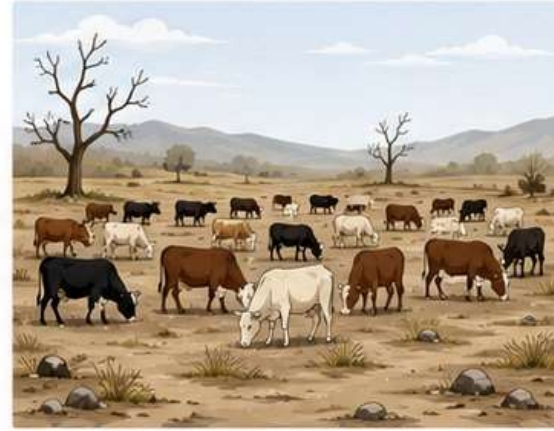
A common pasture available to everyone.

2 Individual incentive



Each person adds more cattle to gain more benefit.

3 Overuse



The pasture becomes crowded and depleted.

4 Collective loss



Everyone ends up worse off as the resource is damaged.

Also applies to other commons



Fisheries
Catching more fish than the population can replace.



Forests
Harvesting more timber than the forest can regenerate.



Groundwater
Pumping more water than aquifers can recharge.



Key idea

Short-term individual gain can create long-term shared loss unless rules, cooperation, or regulation protect the commons.

Three Additional Principles of Sustainability

Economics, political science, and ethics as guides for sustainable decision-making

Full-cost Pricing

from Economics



Prices should reflect hidden environmental and health costs.

Consumers see the true cost of goods and services.

Win-win Solutions

from Political Science



Cooperation and compromise can solve environmental problems.

Benefits are shared by people and the environment.

Responsibility to Future Generations

from Ethics



We should leave life-support systems as good as, or better than, we inherited.

Future people and other species are protected.

Sustainability requires better prices, cooperative politics, and ethical responsibility.

Ecological footprint: “How much nature do we use?”

*Ecological footprint means the amount of biologically productive **land and water** needed to provide the resources a person, community, or country uses and to **absorb the waste** it produces.*

EXAMPLE 1: HIGHER ECOLOGICAL FOOTPRINT



Person who eats a lot of meat, drives daily, uses air conditioning heavily, and buys many new products.

WHAT THEY USE



Meat-rich diet



Drives daily



Heavy use of air conditioning



Buys many new products



Large home, high energy use

LAND AND WATER NEEDED TO SUPPORT THIS LIFESTYLE



Grazing land for meat and dairy



Cropland to grow food and feed



Forest for timber, paper and products



Energy land to produce fuel



Land and water to absorb waste (e.g., CO₂)

RESULT:
LARGER
ECOLOGICAL
FOOTPRINT

=



Needs more land and water than the Earth can renew easily.

EXAMPLE 2: LOWER ECOLOGICAL FOOTPRINT



Person who uses public transport, eats more plant-based food, saves energy, and reuses products.

WHAT THEY USE



More plant-based diet



Uses public transport



Saves energy (e.g., LED lights)



Reuses and buys less



Smaller home, low energy use

LAND AND WATER NEEDED TO SUPPORT THIS LIFESTYLE



Less grazing land needed



Less cropland needed



Less forest area needed



Less energy land needed



Less land and water to absorb waste (e.g., CO₂)

RESULT:
SMALLER
ECOLOGICAL
FOOTPRINT

=



Needs less land and water; easier for the Earth to renew.

Footprint Calculator



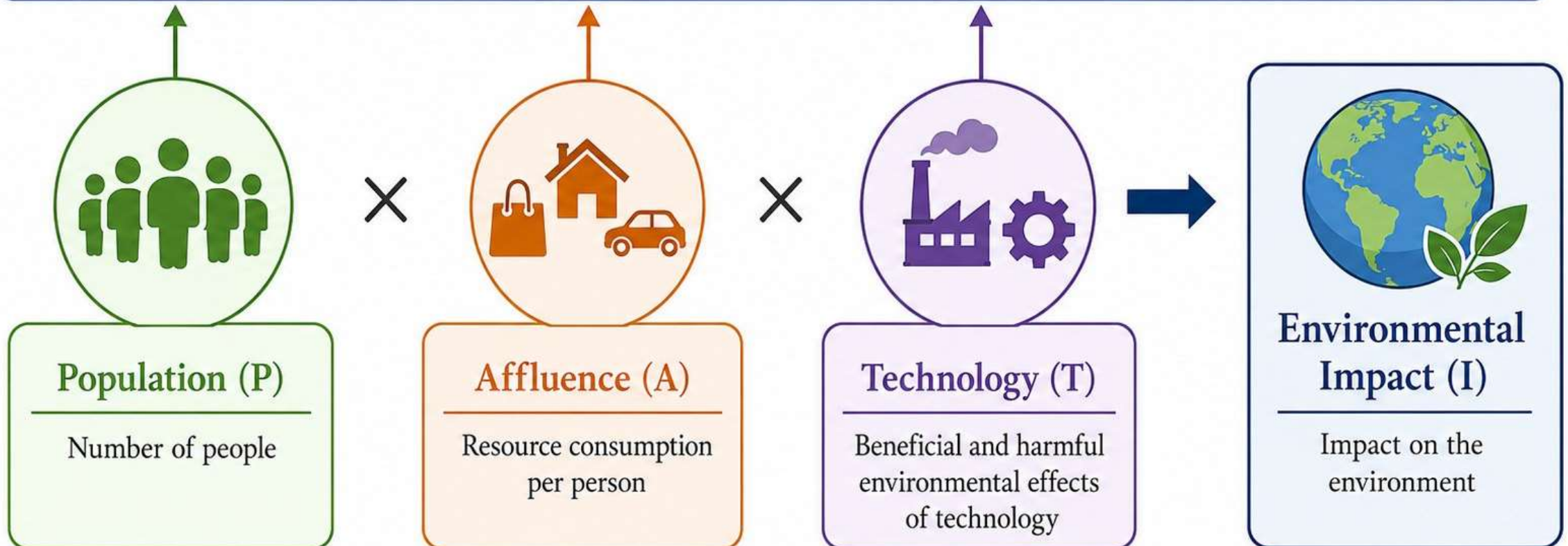
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Our ecological footprint (environmental impact) is increasing

IPAT Model

A simple way to understand environmental impact

$$\text{Environmental Impact (I)} = \text{Population (P)} \times \text{Affluence (A)} \times \text{Technology (T)}$$



Earth Overshoot Day 2026 — and who is actually overshooting

30 July 2026: humanity had used a full year of Earth’s regeneration.

We are consuming nature 73% faster than it regenerates — 1.73 Earths. The highest overshoot ever recorded. Carbon alone is 61% of the footprint.

If everyone lived like... (Country Overshoot Day 2026)

Qatar	4 February	10.0 Earths
United States	14 March	5.1 Earths
China	27 May	2.5 Earths
WORLD AVERAGE	30 July	1.73 Earths
Indonesia	18 October	1.3 Earths
India	— none —	0.75 Earths
Bangladesh	— none —	0.46 Earths

The uncomfortable part

Bangladesh already lives inside one Earth. It has no overshoot day at all.

Bangladesh is also one of the most climate-vulnerable countries in the world.

So: who causes environmental problems, and who pays for them?

Careful with the good news: a small footprint here is mostly a symptom of low income, not of good policy. The question is whether Bangladesh can grow without buying the Qatari footprint along with it.

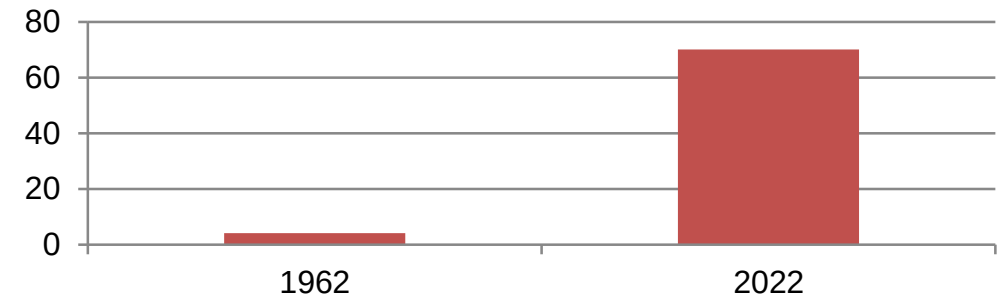


What changes as incomes rise?

Common patterns:

- More appliances + electronics
- More meat/dairy
- More cars + air travel
- More packaging + waste

Example: meat supply per person
(China, kg/person/year)

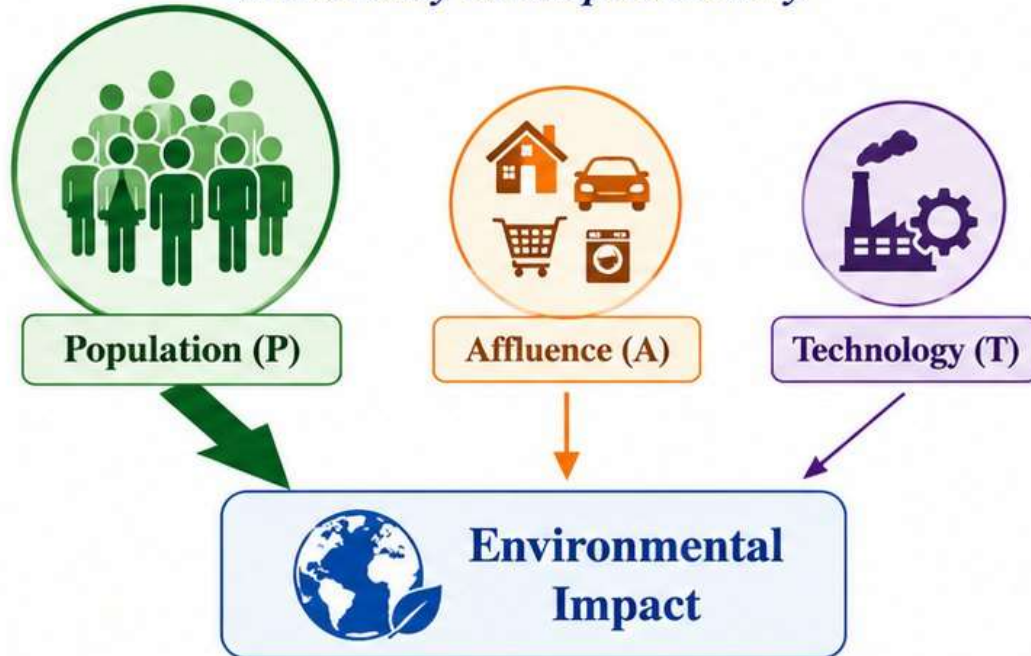


Two IPAT Cases: India and the United States

How different factors can shape environmental impact

Case 1: India

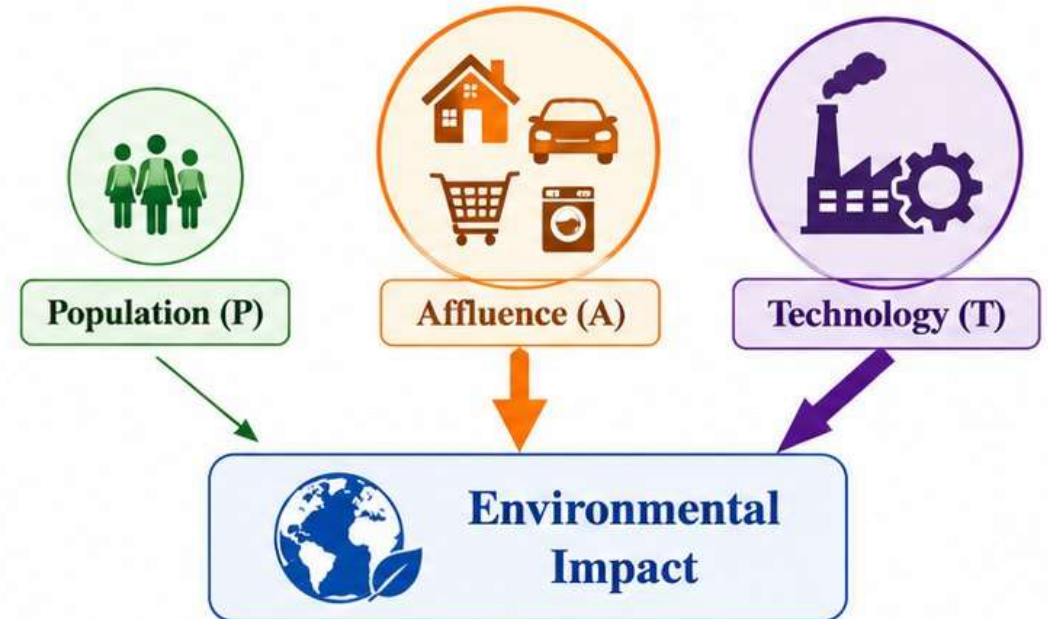
Moderately developed country



Population size is a more important factor than affluence in shaping environmental impact.

Case 2: United States

Highly developed country



Per-person resource use and technology play key roles in shaping environmental impact.



Key idea: Environmental impact depends on the balance among population, consumption, and technology.

Fast Fashion through the IPAT Model

$$\text{Environmental Impact (I)} = \text{Population (P)} \times \text{Affluence (A)} \times \text{Technology (T)}$$

Population (P)

More consumers



- Large global market
- Rising demand for trendy clothing

Affluence (A)

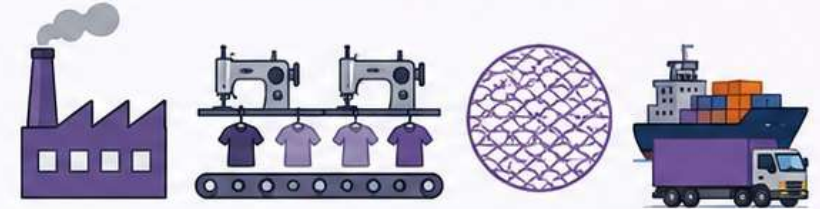
High consumption per person



- Buys more clothes
- Frequent trends
- Short use before disposal

Technology (T)

Mass-production systems



- Cheap synthetic fibers
- Dyeing and finishing
- Global transport

Environmental Impact of Fast Fashion



Textile
waste



Water
pollution



High
energy use



Carbon
emissions



Fast fashion increases environmental impact when many people buy large amounts of cheaply produced clothing.

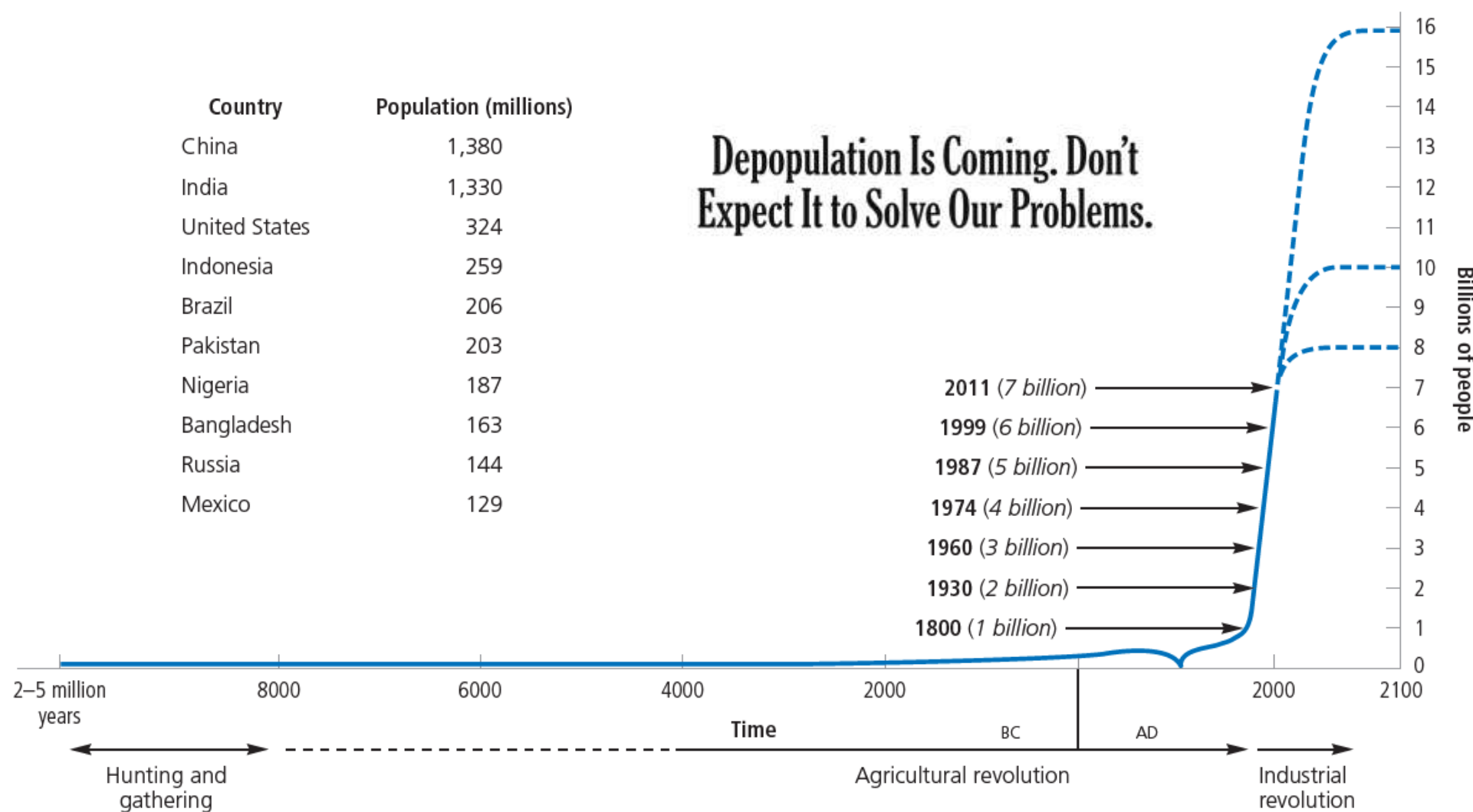
Ghana becomes dumping ground for the world's unwanted used clothes



<https://youtu.be/MHnDqelUh-4?si=fo5BeEoTnJr73NRH>

What Causes Environmental Problems?

Basic causes of environmental problems are (i) population growth, (ii) wasteful and unsustainable resource use, (iii) poverty, (iv) avoidance of full-cost pricing, and (v) increasing isolation from nature (environmental worldview).



What Causes Environmental Problems?

Basic causes of environmental problems are (i) population growth, (ii) wasteful and unsustainable resource use, (iii) poverty, (iv) avoidance of full-cost pricing, and (v) increasing isolation from nature (environmental worldview).

Assumptions and values about how the natural world works and how people should interact with it

1. Human-Centered



- Nature as a support system for humans
- Humans are separate from and in charge of nature
- Earth should be managed for human benefit



Planetary
Management



Stewardship

2. Life-Centered



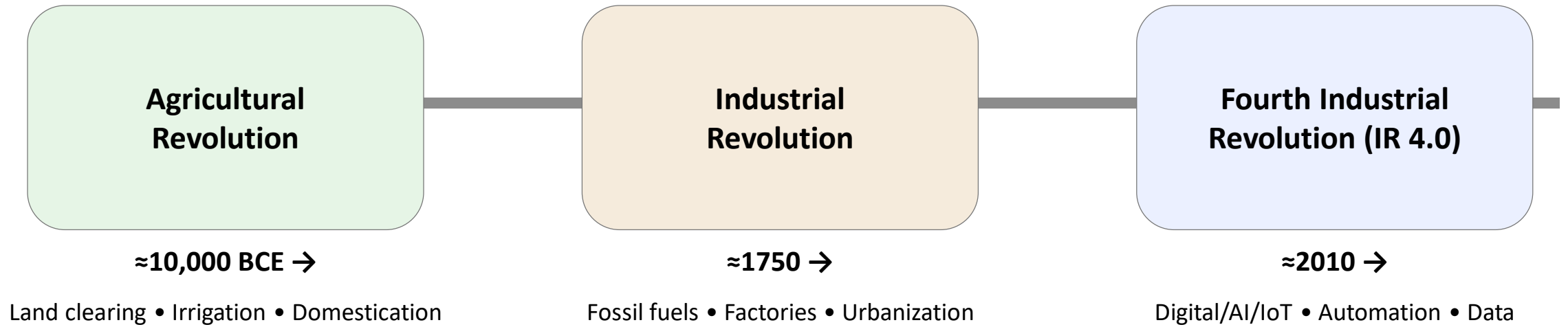
- All species have value
- Each species has an ecological role
- Value is not limited to usefulness to humans

3. Earth-Centered



- People are part of and dependent on nature
- Earth's natural capital exists for all species
- Humans are not the only focus

Big Revolutions & the Environment (Very Quick History)



Which revolution caused the biggest environmental change? Why?

1) Agricultural Revolution → Changing Land & Water

- **Shift:** hunting/gathering → farming + permanent settlements
- **Environmental impacts:** deforestation, habitat loss, soil erosion, irrigation salinity, methane from rice & livestock
- **Also enabled:** stable food supply → population growth → more resource demand

2) Industrial Revolution + IR 4.0 → Energy, Emissions, and ‘Smart’ Solutions

- **Industrial Revolution:** coal/oil + machines → mass production and rapid urban growth
- **Impacts:** air pollution, CO₂ rise, mining impacts, toxic wastes, river contamination; “take-make-waste” economy
- **IR 4.0 (today):** AI, sensors, drones, big data → can cut waste (optimization) BUT can increase e-waste + electricity demand (data centers)
- **Key idea:** Technology helps most when policy + incentives + behavior also change.

The newest commons: electricity, chips and the cloud

What the IEA reported in April 2026

- Data-centre electricity demand grew 17% in 2025 — while total global electricity demand grew 3%
- Data-centre consumption is set to double by 2030; AI-focused consumption is set to triple
- Five big technology firms spent over \$400 billion on capacity in 2025, and expect to spend ~75% more in 2026
- Those same firms signed roughly 40% of the world's corporate renewable power deals in 2025
- Efficiency per AI task keeps improving — and total consumption keeps rising anyway

Put it through IPAT

$$I = P \times A \times T$$

T got better: each query costs less energy than it did two years ago.

A got much bigger: we all ask far more of it, far more often.

If T improves and I still rises, which term is winning?

Where does the electricity come from — and where does the cooling water come from?

Same structure as the pasture, the river and the fishery: a shared resource, individual incentive to take more, collective loss.

Back to Gazipur — would you change your vote?

Where the world actually stands, as of 2026:

1.44 °C

above pre-industrial in 2025
(WMO, January 2026)

2023–25

the three warmest years on record
in all eight global datasets

Top 10

Dhaka appeared repeatedly among
the world's 10 most polluted cities in 2026

Three things that change the Gazipur answer

- Full-cost pricing — if the canal had a price, Option 1 was never the cheap option
- Rules for the commons — no single factory can clean a river alone
- The time horizon — the clean line paid back in under three months

Every environmental problem in this course is a decision someone made with the wrong price, the wrong rules, or the wrong time horizon.

Thank you